

Pharmacy Management System V12

Termux • Node.js • SAS • LaTeX Integration

System Design Document

Inventory • Sales • Analytics • Reporting • Backup • Health
Monitoring

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1 Executive Summary

Overview

The Pharmacy Management System V12 is a lightweight, offline-ready pharmacy platform designed for Termux environments. It integrates inventory management, sales processing, analytics, SAS export, backups, and LaTeX reporting into a single unified system.

2 System Architecture

The system consists of modular components:

- Inventory Engine
- Sales Engine
- Analytics Engine
- Reporting Engine
- Web Dashboard
- SAS Export Engine
- Backup System
- Health Monitoring

3 Operational Menu Structure

Option	Function
1	Add Medicine
2	List Medicines
3	Sell Medicine
4	Search Medicine
5	Alerts
6	Analytics
7	Web Dashboard
8	Reports (TXT)
9	Stop Server
10	Exit
11	SAS Export
12	Backup System
13	System Health Check

4 System Highlights

Key Features

- Fully offline Termux-compatible system
- Lightweight JSON-based storage
- Web dashboard for real-time monitoring
- SAS-compatible data export (CSV)
- Automated backup system
- Audit logging for all actions
- Health and inventory monitoring
- TXT + LaTeX reporting support
- Modular and extendable architecture

5 Reporting System

5.1 TXT Reports

Generated for quick operational summaries:

```
reports/pharmacy_report.txt
```

5.2 LaTeX Reports

Professional PDF output using:

```
~/completex1.sh
```

6 SAS Integration

Exports structured datasets for analytics:

```
exports/medicines_sas.csv  
exports/sales_sas.csv
```

Compatible with SAS, Python, R, and Excel.

7 Backup System

Creates timestamped backups of:

- Medicines database

- Sales records
- Audit logs

Ensures full data recovery capability.

8 System Health Check

Provides system diagnostics:

- Total medicines
- Total sales
- Revenue summary
- Low stock alerts

9 Future Additions (Roadmap V13+)

System Evolution Roadmap

The Pharmacy Management System is designed as an evolving platform. Future versions will extend its intelligence, visualization, and clinical utility.

9.1 Advanced Analytics and AI

- Predictive demand forecasting for medicines
- Seasonal inventory optimization
- AI-based sales trend detection
- Early warning system for stock depletion

9.2 Visualization and Reporting Enhancements

- Embedded charts using `pgfplots`
- Revenue trend graphs inside PDF reports
- Interactive-style dashboard reports (static PDF equivalent)
- Pharmacy performance scorecards

9.3 Clinical and Pharmacy Intelligence

- Drug interaction warning layer
- Expiry risk prediction system
- High-risk medicine monitoring
- Prescription pattern analysis

9.4 System Expansion

- Multi-branch pharmacy support
- Role-based access (Admin / Pharmacist / Cashier)
- User authentication system
- Cloud synchronization option

9.5 Automation and Integration

- Auto-generated daily/weekly reports
- Scheduled SAS exports
- Automatic backup scheduling
- WhatsApp / SMS notification alerts (future plugin)

9.6 User Experience Improvements

- Enhanced CLI interface with structured panels
- Improved web dashboard UI
- Mobile-optimized dashboard view
- Barcode / QR code scanning support

10 Conclusion

Pharmacy Management System V12 delivers a complete lightweight pharmacy platform combining operational management, analytics, reporting, SAS interoperability, and LaTeX documentation within a single Termux-based ecosystem.